



Hoggar Trail

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Difficulty Level : Hard

Hoggar Trail

The Hoggar Mountains, located in southern Algeria near Tamanrasset, are among the most famous mountainous regions in the country. A hiking club recorded the altitude of n checkpoints along a long desert trail, where the i -th checkpoint has altitude a_i .

A hiker may choose any subsequence of these checkpoints, keeping their original order, to form a personal route. The **effort** of a route is defined as the total altitude change between consecutive chosen checkpoints. More formally, if the chosen route is $s_1, s_2, \dots, s_k$ then its effort is:

$$\sum_{i=2}^k |s_i - s_{i-1}|$$

A route containing fewer than two checkpoints has effort 0.

Task: Compute the total sum of efforts over all possible routes, modulo $10^9 + 7$.

Input

The first line contains a single integer n ($1 \leq n \leq 10^5$), the number of checkpoints.

The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$), where a_i is the altitude of the i -th checkpoint.

Output

Print a single integer — the sum of efforts of all possible routes, modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^5$
- $-10^9 \leq a_i \leq 10^9$
- Time limit: **2 seconds**
- Memory limit: **256 MB**

Sample Input 1

3 2 7 5

Sample Output 1

17

Sample Input 2

1 5

Sample Output 2

0

Explanation

For the array $[2, 7, 5]$, we consider all subsequences with at least two elements and sum their efforts:

- $[2, 7] \Rightarrow |7 - 2| = 5$
- $[2, 5] \Rightarrow |5 - 2| = 3$
- $[7, 5] \Rightarrow |5 - 7| = 2$
- $[2, 7, 5] \Rightarrow |7 - 2| + |5 - 7| = 5 + 2 = 7$

Total: $5 + 3 + 2 + 7 = 17$.

For a single checkpoint, no route with two or more elements exists, so the answer is 0.